King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 201 Major Exam II Term 243

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. If $x \ln y + y^2 z + z^2 = 8$, then $\dfrac{\partial z}{\partial x} =$

(a) $-\dfrac{y^2 z + z^2}{y^2 + 2z}$

(b) $\dfrac{x}{y^2 + 2z}$

(c) $\dfrac{-\ln y}{y^2 + 2z}$

(d) $-\dfrac{x \ln y}{y + 2z}$

(e) $-\dfrac{x \ln y}{y^2 + z^2 - 8}$

2. If $w = ze^{xy}$, $x = s - t$, $y = s + t$, $z = st$, then $w_s =$

(a) $(1 + s^2)e^{s^2 - t^2}$

(b) $e^{s^2 - t^2}$

(c) $t(1 - s^2)e^{s^2 - t^2}$

(d) $t(1 + 2s^2)e^{s^2 - t^2}$

(e) $2ste^{s^2 - t^2}$

3. Let

$$f(x, y) = |xy| \left(\frac{x^2 - y^2}{x^2 + y^2} \right).$$

The value of $f(0, 0)$ such that f is continuous at the origin is

(a) -2

(b) -1

(c) 2

(d) 0

(e) 1

4. $\lim_{(x,y)\to(1,1)} \frac{2x-y-1}{\sqrt{2x-y}-1} =$

(a) DNE

(b) 1

(c) 0

(d) -2

(e) 2

5. The line through the point $(-4, 5, 2)$ and perpendicular to the plane given by $-x + 2y + z = 5$ also passes through the point $(-3, a, b)$. Then $a + b =$

(a) 7

(b) 12

(c) 10

(d) 4

(e) 11

6. The graph of the equation

$$2x^2 + 4y^2 + 2z^2 - 8x + 8y - 4z = c$$

is an ellipsoid when $c =$

(a) -19

(b) -15

(c) -21

(d) -17

(e) -11

7. The distance between the point $(-2, 1, 3)$ and the line given by

$$x = 1 - t, \quad y = 2 + t, \quad z = -2t$$

is equal to

(a) $\frac{2}{\sqrt{3}}$

(b) $\frac{5}{\sqrt{3}}$

(c) $\sqrt{3}$

(d) $\frac{7}{\sqrt{3}}$

(e) $\frac{1}{\sqrt{3}}$

8. The plane $ax + by + cz = 21$ contains the points $(4, 1, 3)$ and $(2, -2, 1)$ and is perpendicular to the plane $x + 2y - z = 2$. Then $a + b + c =$

(a) 5

(b) 4

(c) 7

(d) 3

(e) 2

9. If $z = \arctan\left(\frac{y}{x}\right)$, then

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} =$$

(a) $\dfrac{1}{(x^2+y^2)^2}$

(b) $-\dfrac{x+y}{(x^2+y^2)^2}$

(c) $\dfrac{x^2-y^2}{(x^2+y^2)^2}$

(d) 0

(e) $\dfrac{2xy}{(x^2+y^2)^2}$

10. The level surfaces of the function $f(x, y, z) = 4x^2 + 4y^2 - z^2$ are

(a) elliptic paraboloids

(b) cones only

(c) ellipsoids

(d) hyperboloids of one sheet only

(e) hyperboloids of one sheet, hyperboloids of two sheets, and cones

11. Let $w = \sqrt{x-y} + \sqrt{y-z}$. The value of $\dfrac{\partial w}{\partial x} + \dfrac{\partial w}{\partial y} + \dfrac{\partial w}{\partial z}$ at $x = 5$, $y = 3$ and $z = 1$ is

(a) 0

(b) $\dfrac{3}{\sqrt{2}}$

(c) $-4\sqrt{2}$

(d) $\sqrt{2}$

(e) $\dfrac{1}{\sqrt{2}}$

12. Let D be the domain and R be the range of $f(x,y) = \arccos(x-y)$. Then

(a) $D = \{(x,y) : \ -\frac{\pi}{2} \leq x - y \leq \frac{\pi}{2}\}$ and $R = [0, \infty)$

(b) $D = \{(x,y) : \ -x \leq y \leq 1 - x\}$ and $R = [-\pi/2, \pi/2]$

(c) $D = \{(x,y) : \ -\infty < x < \infty$ and $-\infty < y < \infty\}$ and $R = [-1, 1]$

(d) $D = \{(x,y) : \ -1 - x \leq y \leq 1 + x\}$ and $R = [0, \pi]$

(e) $D = \{(x,y) : \ -1 + x \leq y \leq 1 + x\}$ and $R = [0, \pi]$

13. Using differentials, the quantity

$$(2.01)^2(6.01) - 2^2 \cdot 6$$

is approximately equal to

(a) 0.28

(b) 0.30

(c) 0.24

(d) 0.26

(e) 0.14

14. Let $w = f(x,y)$ with $x = g(s,t)$ and $y = h(s,t)$, where f, g, and h are differentiable functions, with the following values:

$g(1,2)$	$h(1,2)$	$g_s(1,2)$	$h_s(1,2)$	$f_x(4,3)$	$f_y(4,3)$
4	3	-3	5	-4	6

Then $w_s(1,2) =$

(a) 20

(b) 42

(c) -39

(d) 36

(e) -38

15. The distance between the skew lines

$$L_1 : \ x = 2t, \ y = 4t, \ z = 6t \qquad \text{and} \qquad L_2 : \ x = 1 - s, \ y = 4 + s, \ z = -1 + s$$

is equal to

(a) $\dfrac{20}{\sqrt{26}}$

(b) $\dfrac{10}{\sqrt{13}}$

(c) $2\sqrt{13}$

(d) $\dfrac{5\sqrt{26}}{13}$

(e) $\sqrt{26}$

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	$F_x = \ln y$, $F_z = y^2 + 2z$, so $z_x = -F_x/F_z = \frac{-\ln y}{y^2+2z}$.
2	**(d)**	$w = st\,e^{s^2-t^2}$, so $w_s = te^{s^2-t^2}(1+2s^2)$.
3	**(d)**	$\lvert f\rvert \le \lvert xy\rvert \to 0$ by squeeze, so $f(0,0) = 0$.
4	**(e)**	Factor: ratio $= \sqrt{2x-y}+1 \to 2$.
5	**(d)**	Direction $\langle -1, 2, 1\rangle$; $x = -3 \Rightarrow t = -1$, so $(a, b) = (3, 1)$ and $a + b = 4$.
6	**(e)**	Completing squares: right side $c + 14$ must be positive, so $c > -14$: $c = -11$.
7	**(d)**	$D = \frac{\lvert\overrightarrow{QP}\times\mathbf{u}\rvert}{\lvert\mathbf{u}\rvert} = \frac{7\sqrt{2}}{\sqrt{6}} = \frac{7}{\sqrt{3}}$.
8	**(e)**	Solve $4a + b + 3c = 21$, $2a - 2b + c = 21$, $a + 2b - c = 0$: $(a, b, c) = (7, -4, -1)$, sum 2.
9	**(d)**	$z_x = \frac{-y}{x^2+y^2}$, $z_y = \frac{x}{x^2+y^2}$; $z_{xx} = -z_{yy}$, so the sum is 0.
10	**(e)**	$k > 0$: one sheet; $k = 0$: cone; $k < 0$: two sheets.
11	**(a)**	$w_x + w_y + w_z = 0$: the terms $\pm\frac{1}{2\sqrt{x-y}}$ and $\pm\frac{1}{2\sqrt{y-z}}$ cancel.
12	**(e)**	$-1 \le x - y \le 1 \iff x - 1 \le y \le x + 1$; range of arccos is $[0, \pi]$.
13	**(a)**	$f = x^2y$: $df = 2xy\,dx + x^2\,dy = 24(0.01) + 4(0.01) = 0.28$.
14	**(b)**	$w_s = f_x g_s + f_y h_s = (-4)(-3) + (6)(5) = 42$.
15	**(a)**	$\mathbf{u}_1 \times \mathbf{u}_2 = \langle -2, -8, 6\rangle$; $D = \frac{40}{2\sqrt{26}} = \frac{20}{\sqrt{26}}$.